Course Syllabus

MATH 5101 - Analysis I

Instructor Name: Eugene Tang

Email: e.tang@northeastern.edu (mailto:e.tang@northeastern.edu) — Please include "Math 5101" in the subject line to prevent any emails from going to spam!

Office Hours: 437 Lake Hall -- Tuesdays directly before and after class (2-3:20 pm and 5:10-7 pm).

Note: *If I am not in my office, you can try the math lounge upstairs from my office.*

Lecture Time/Location: Tuesday, Friday from 3:25 – 5:05 pm — 310 Hurtig Hall

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Note that this Syllabus may be subject to change. It is *your responsibility* to be aware of any changes the instructor may make to the syllabus as they are announced on Canvas/email. Please read all emails and announcements.

About this Course and its Prerequisites: MATH 5101 is a graduate level real analysis course. While the contents are largely self-contained, this course is typically ***not*** meant to be a first introduction to real analysis. The relation between this course and an undergraduate real analysis course is that we cover largely the same content but at a greater level of abstraction and generality. If you already have a very strong background in real analysis at the undergraduate level, you may want to consider MATH 5102, which is the follow-up course covering measure theory.

Soft prerequisites (things you should know, but can be picked up during the course):

1. Mathematical maturity. This course expects you to come in with an understanding of how to read and write proofs and follow mathematical reasoning.

2. Basic notions of set theory. You should understand basic operations on sets (e.g. unions, intersections, power sets, etc), as well as basic notions of cardinality (countable vs. uncountable).

3. Basic notions of linear algebra. You should be familiar with the basic properties of finite dimensional vector spaces over the real numbers, as well as linear maps between them.

4. Basic notions of topology. We will cover the required topology starting from scratch, but the pace of the course means that it will not be a leisurely introduction. The topology covered in this course is really meant more to enrich than to introduce, so existing familiarity with topology is highly recommended. If you have no topology background, I highly recommend the textbook "*Topology*" by Munkres. It might be beneficial to through as much of Chapter 2 as possible (especially Sections 12 to 18).

Textbook and Course Contents: We will follow our own lecture notes, which will be posted weekly.

In terms of actual textbooks, there are two recommendations and one supplement. The first is Rudin's "*Principles of Mathematical Analysis*". Rudin's text is an undergraduate analysis textbook. It is terse and very efficient at building the structural core of real analysis. Our course will roughly follow the logical progression of Rudin's text, but often go beyond it in depth.

A second book, "Real Mathematical Analysis" by Pugh, is the closest in spirit to the lecture notes. It is a relatively newer book, but already quickly establishing itself as a classic; often being touted as a modernized version of Rudin's text. It is somewhat idiosyncratic in its notation, terminology, and style, but much better at presenting the intuition and the "big picture" compared to Rudin.

A supplementary text will be "*Topology*" by Munkres, for all your point-set topology needs. The first half in particular will contain all the topology we need in this course and more.

We will aim to cover the majority of the first nine chapters of Rudin (roughly the first 5 chapters of Pugh). The contents of the course are approximately as follows:

1. Archimedean Ordered Fields and the Real Numbers.

2. Topological and Metric Spaces.

3. Integration and Differentiation on $\mathbb{R}$.

4. Function spaces and uniform convergence.

Homework Assignments: There will be approximately ***5-7 homework assignments***, roughly one every two weeks. You are allowed, and encouraged, to work on the assignment problems together with your classmates. If you do so, please acknowledge the people you have collaborated with on your assignment. The final write up and submission for each assignment **must be done individually**.

Assignments are to be submitted in PDF format typeset using Latex.

Late Policy: You will be allowed one free late submission (up to 48 hours) for your homework assignments. Otherwise, no late assignments will be accepted.

AI Policy: The University's Policy on AI Systems defines an AI system as "an engineered or machine-based system or functionality that is designed, for a given set of objectives, to generate outputs such as text, images, predictions, or to make recommendations or decisions influencing human action, or real or virtual environments. AI Systems are also designed to operate with varying levels of autonomy."

In this course, **the usage of AI systems such as Claude, ChatGPT, or Gemini is prohibited** for homework and assignment purposes.

The use of such systems can be very helpful when learning and reviewing the material, and I encourage its use for these purposes, with the appropriate discretion. However, at some point, you need to work through the logic and problem-solving process yourself. The assignments are designed specifically to give you that opportunity. The use of generative AI technology and other intellectual shortcuts can, and will, impede your ability to truly learn the course material at a deep level. Use them at your own discretion.

Midterm and Final Examinations: There will be no midterm or final exams.

Grading: Your final grade will be based entirely on your assignments. All assignments will be equally weighted.

Letter Grade: Your final letter grade will be based on the following:

A: 93-100	A-: 90-92	B+: 87-89	B: 83-86	B-: 80-82	C+: 77-79
C: 73-76	C-: 70-72	D+: 67-69	D: 63-66	D-: 60-62	F: 0-59

Incomplete Grade: The I grade (incomplete) will be given only rarely. It is intended to cover real emergency situations in which a student who is doing reasonably well (C- or better) is unable, due to circumstances beyond the student's control, to complete all course requirements (e.g., is unable to take the final exam due to hospitalization). An I grade may not be used to rescue a failing grade, or to postpone the final.

Attendance: Attendance is not mandatory and will not be enforced. However, the course moves extremely quickly. It is possible to fall behind in a single day. You are responsible for all information given when you are absent.

Cell phones and Laptops: The use of cell phones is discouraged during class. Please make sure all phones are silenced during class time. The use of laptops are discouraged except for the express purpose of notetaking.

Disability Resource Center: The university's **Disability Resource Center (http://www.northeastern.edu/drc/)** works with students and faculty to provide students who qualify under the Americans with Disabilities Act with accommodations that allow them to participate fully in the activities at the university. Ordinarily, students receiving such accommodations will deliver teacher notification letters at the beginning of the semester. Students have the right to choose whether to disclose their specific disabilities to instructors but must provide a letter to receive accommodation.

Academic Integrity Policy: Cheating will not be tolerated. All incidents of cheating will be reported. From the Academic Integrity Policy: (see: **http://www.northeastern.edu/osccr/academic-integrity-policy/ (http://www.northeastern.edu/osccr/academic-integrity-policy/)**
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"A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues."

Title IX Protections and Resources: Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal funding. Any NU community member who has experienced such discrimination, sexual assault, relationship violence, stalking, coercion, and/or sexual harassment, is encouraged to seek help. Confidential support and guidance can be found through **University Health and Counseling Services (https://www.northeastern.edu/uhcs/)**, the Northeastern **Center for Spirituality, Dialogue, and Service (https://www.northeastern.edu/spirituallife/)**, and the **Office of Prevention and Education at Northeastern (OPEN) (https://studentlife.northeastern.edu/open/)**. Note that faculty members are considered "responsible employees" at Northeastern University, meaning they are required to report all allegations of sex or gender-based discrimination to the Title IX Coordinator. For additional information and assistance please see the **Office of Institutional Diversity and Inclusion (https://provost.northeastern.edu/oidi/)** webpage.

Inclusion and Diversity: I value all students as persons, regardless of their background, country of origin, race, religion, gender, sexual orientation, ethnicity, or disability status, and am committed to providing a climate of excellence and inclusiveness within all aspects of the course. If there are aspects of your culture or identity that you would like to share with me as they relate to your success in this class, I would be happy to discuss these with you. Also, if you have any concerns in this area or are facing any special issues or challenges, I encourage you to discuss the matter with me as you feel comfortable, with assurance of full confidentiality (the only exception being mandatory reporting of NU Academic Integrity Policy violations and Title IX sex and gender discrimination).

We Care and Mental Health

We Care (https://we-care.studentlife.northeastern.edu/) offers supports for students during times of difficulty or challenge. You can find WeCare at 226 Curry Student center Monday - Friday from 8:30-5:00, call at 617-373-7591, or email **wecare@northeastern.edu (mailto:wecare@northeastern.edu)**.

Mental health resources are available through Northeastern's University **Health and Counseling Services (https://uhcs.northeastern.edu/)**. Northeastern has also added **Find@Northeastern (https://uhcs.northeastern.edu/findnortheastern/)**, which is a 24/7 mental health consulting line and can be reached at 1-877-223-9477.

College of Science Policies: The current College of Science Academic Course Policies is available at: **https://cos.northeastern.edu/wp-content/uploads/2012/10/COS-teaching-policies-April-2017.pdf (https://cos.northeastern.edu/wp-content/uploads/2012/10/COS-teaching-policies-April-2017.pdf)**